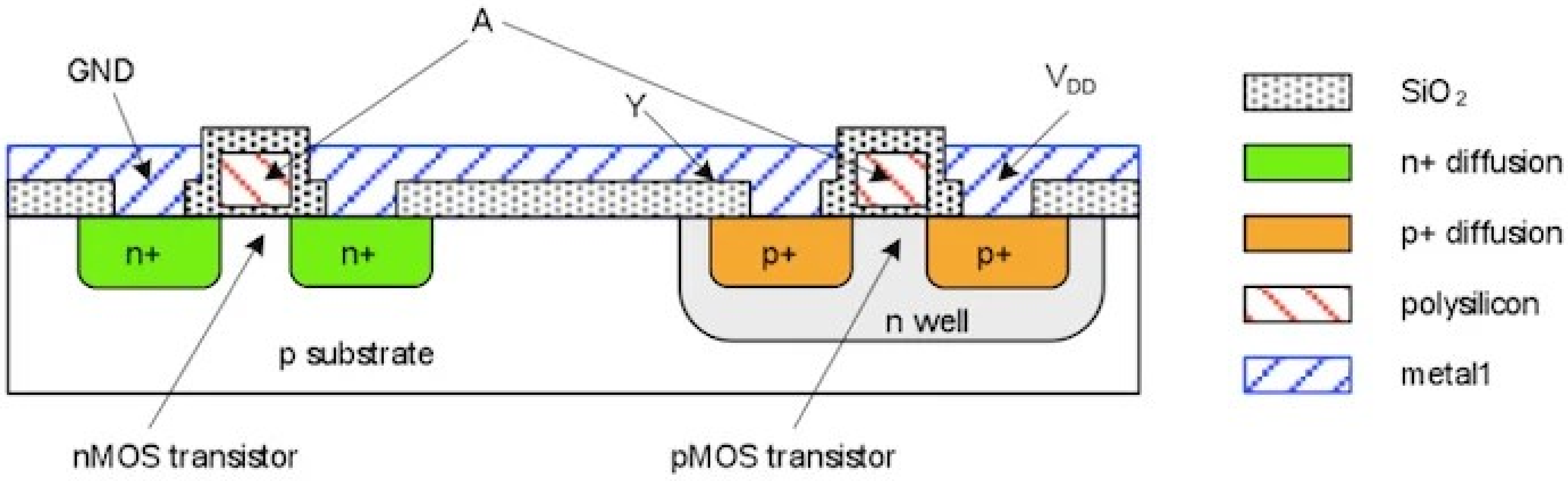
MOS Transistor Types: nMOS & pMOS

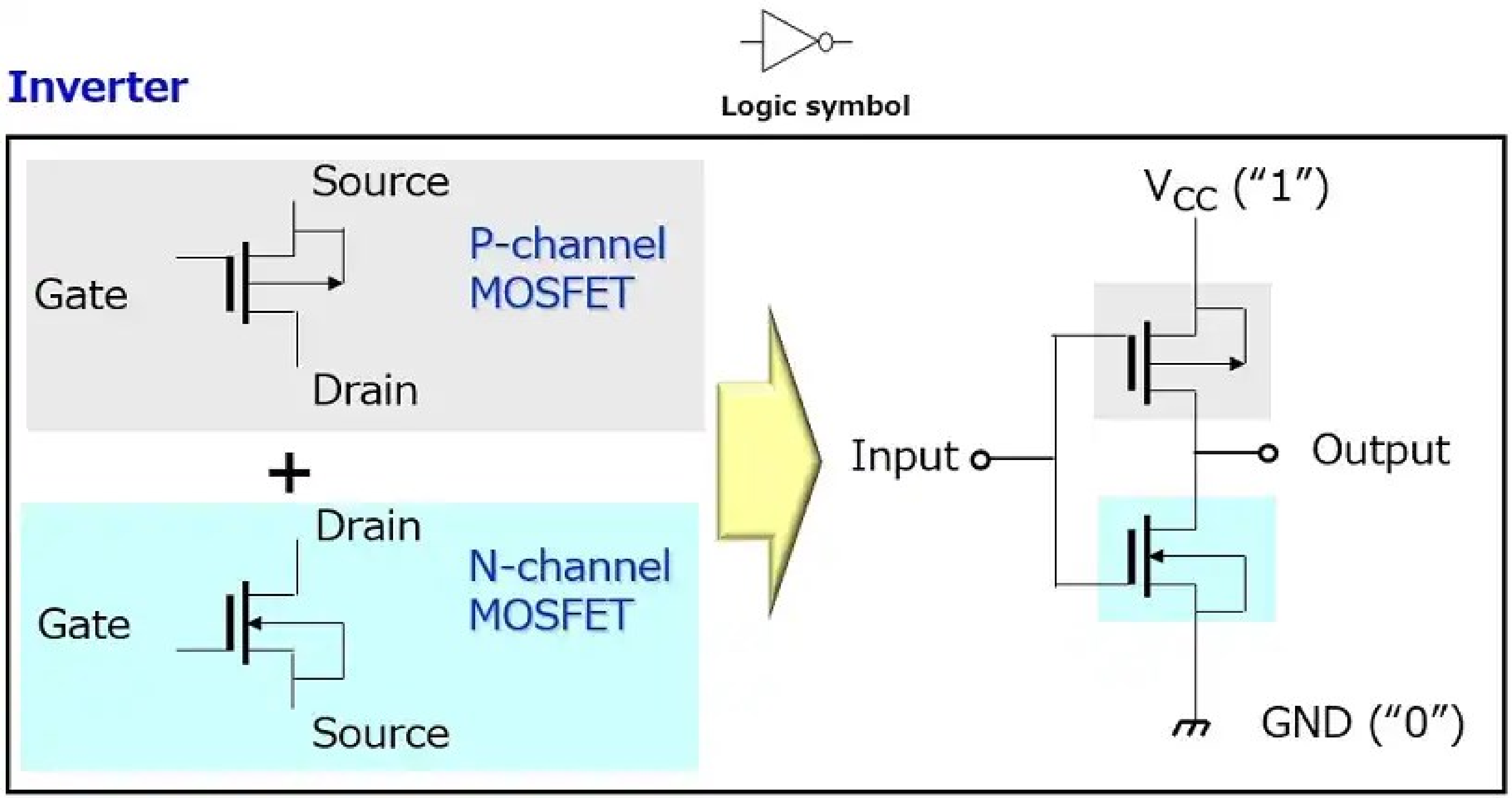
Isolation without Bulk electrode



https://www.allaboutcircuits.com/uploads/articles/NMOS_and_PMOS_silicon_layouts_in_a_CMOS_technology.jpg

CMOS (Complementary Metal-Oxide-Semiconductor)

UNIVERSITY OF PASSAU
Faculty of Computer Science and Mathematics



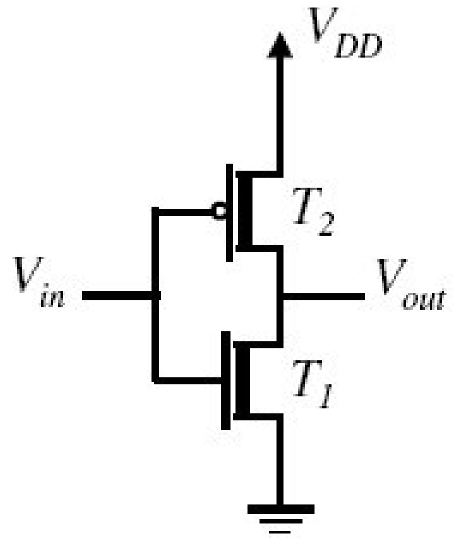
https://toshiba.semicon-storage.com/content/dam/toshiba-ss-v3/master/en/semiconductor/knowledge/e-learning/cmos-logic-basics/chap2-1_en.jpg

CMOS Inverter

Truth Table

V_{in}	T_1	T_2	V_{out}
0	*off*	*on*	1
1	*on*	*off*	0

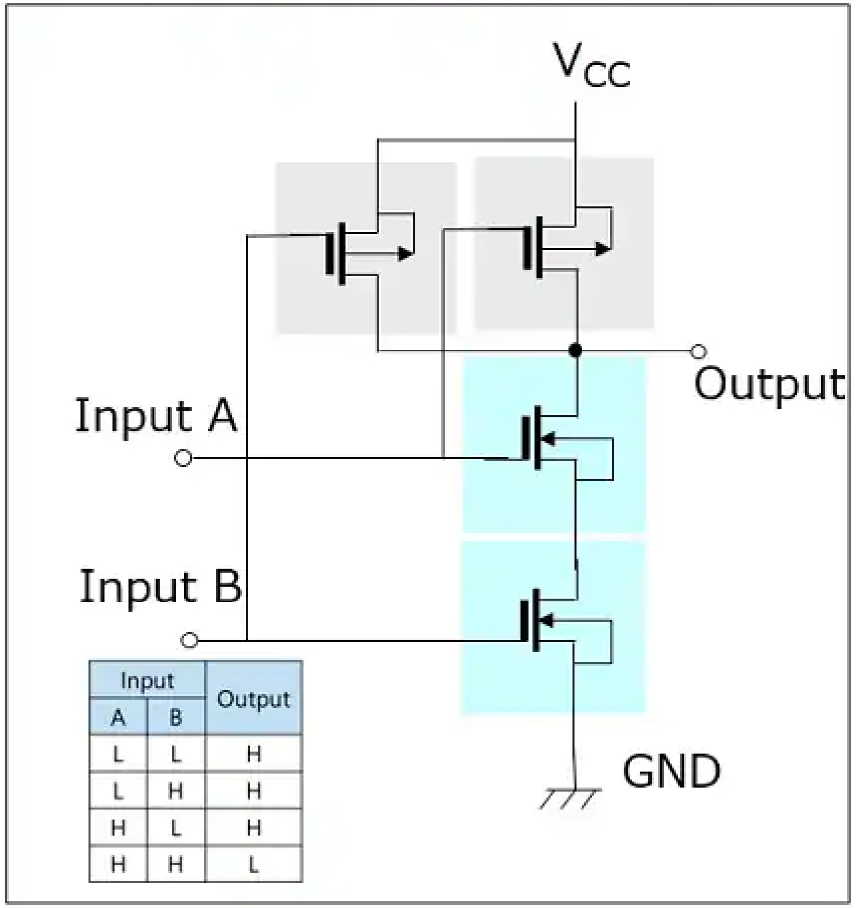
Integrated Circuit



https://www.oreilly.com/api/v2/epubs/9780470900550/files/images/ch005-f007.jpg

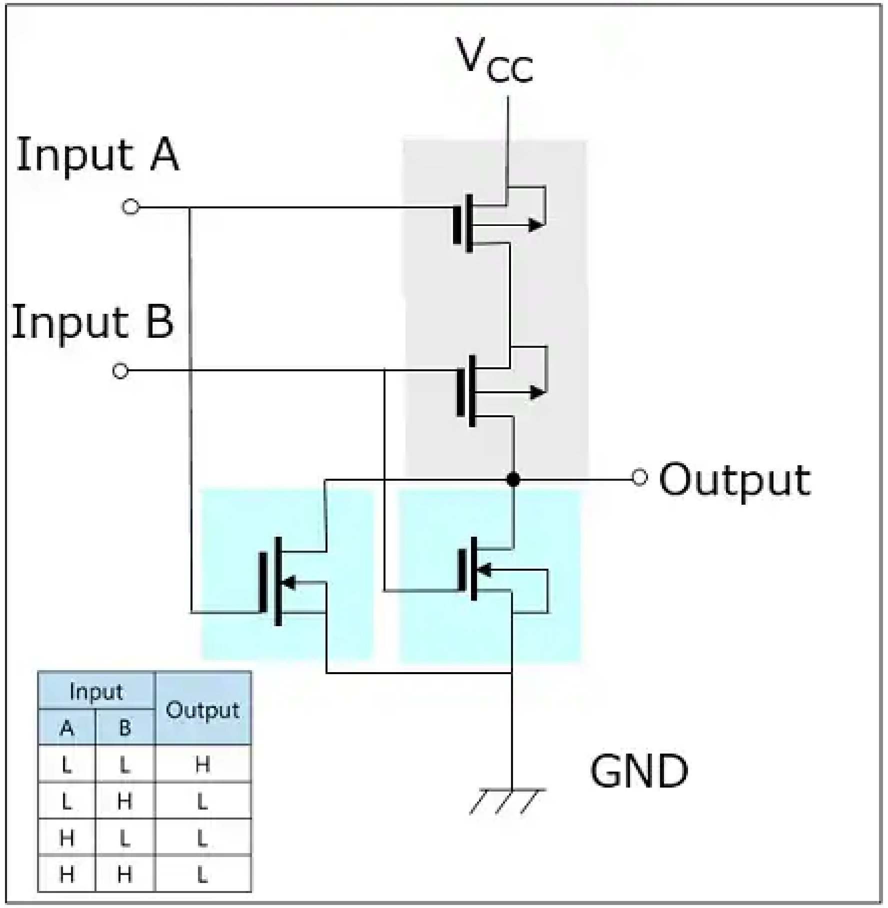
CMOS NAND and NOR gates

2-input NAND gate



Input A	Input B	Output
L	L	H
L	H	H
H	L	H
H	H	L

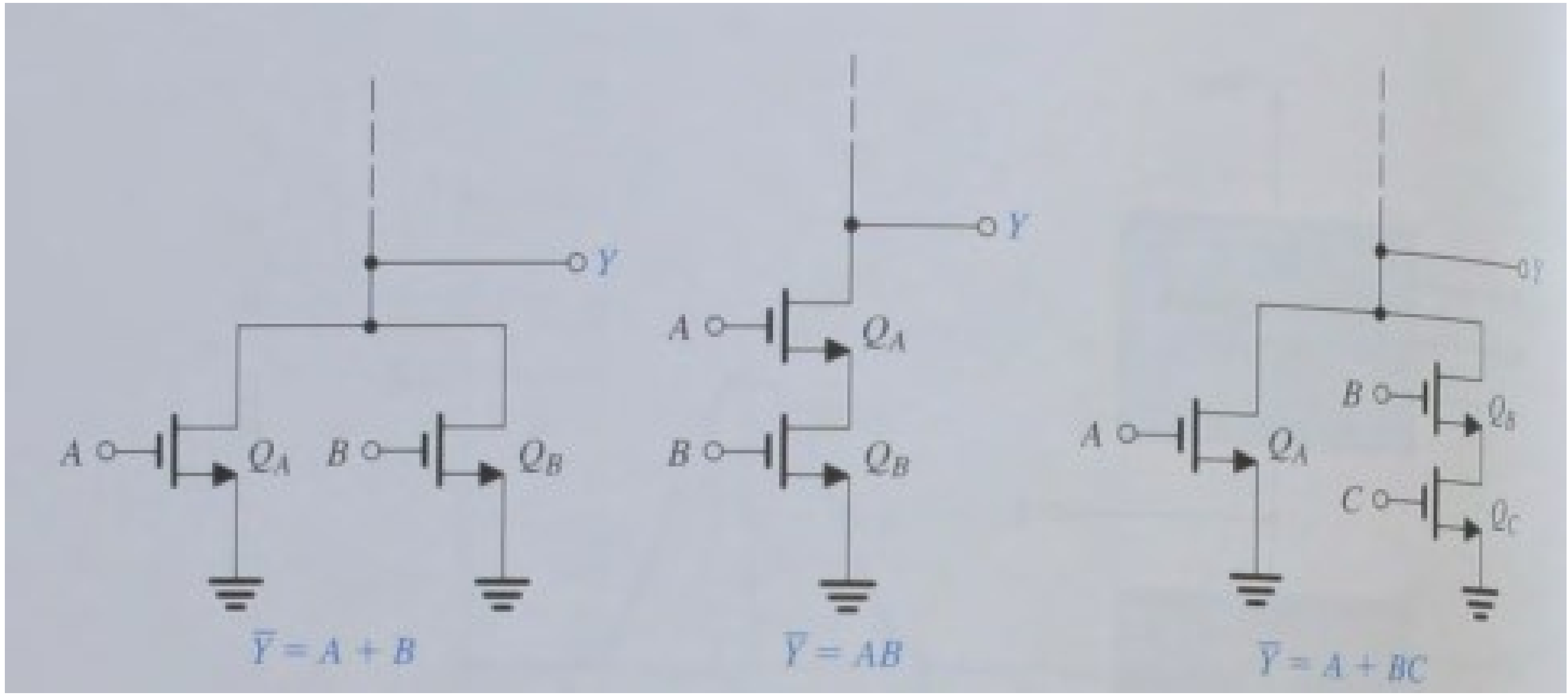
2-input NOR gate



Input A	Input B	Output
L	L	H
L	H	L
H	L	L
H	H	L

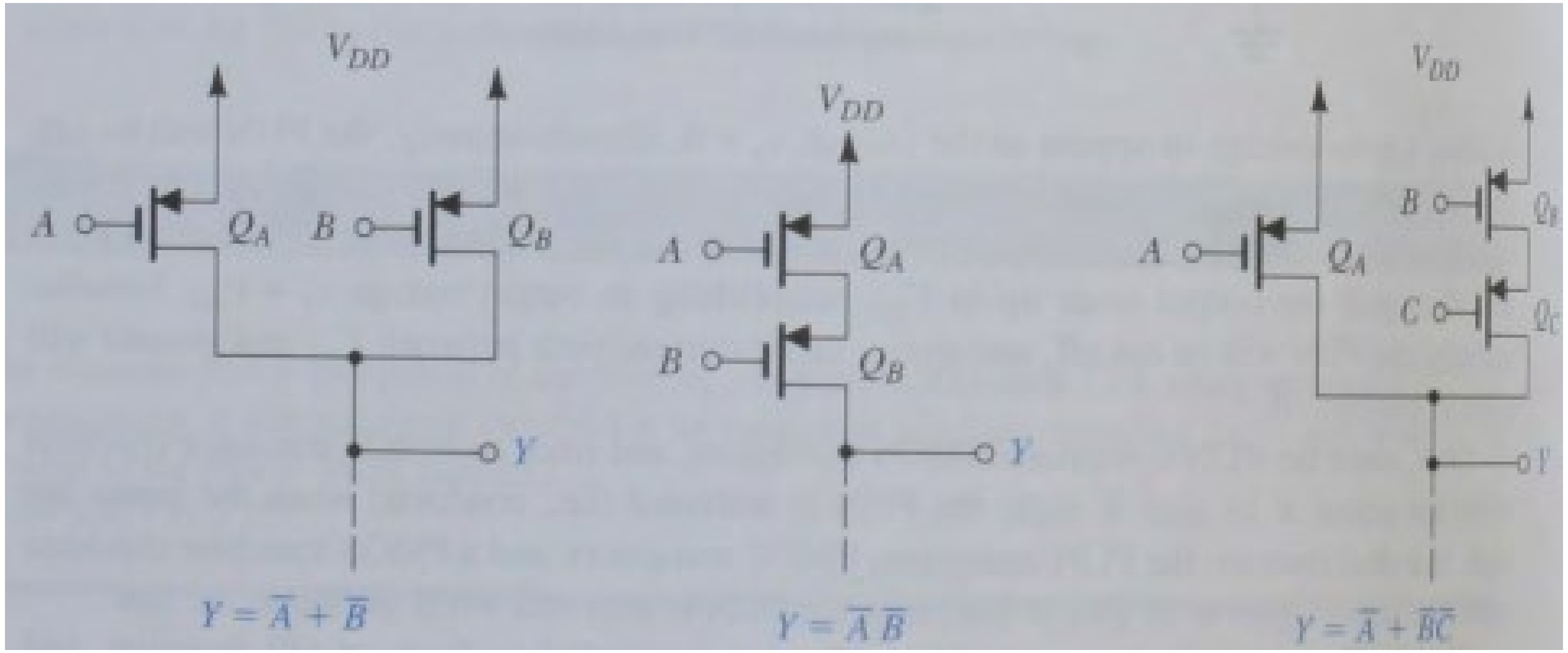
https://toshiba.semicon-storage.com/content/dam/toshiba-ss-v3/master/en/semiconductor/knowledge/e-learning/cmos-logic-basics/chap2-2-3_en.jpg

Applications of CMOS: Digital Circuits



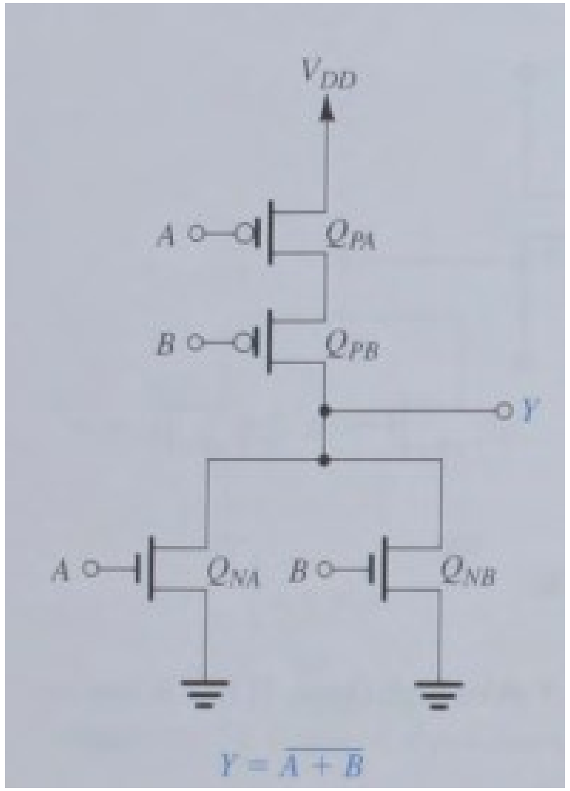
Sedra, Smith, Carusone & Gaudet: “Microelectronic Circuits”, Oxford University Press, 2019.

Applications of CMOS: Digital Circuits



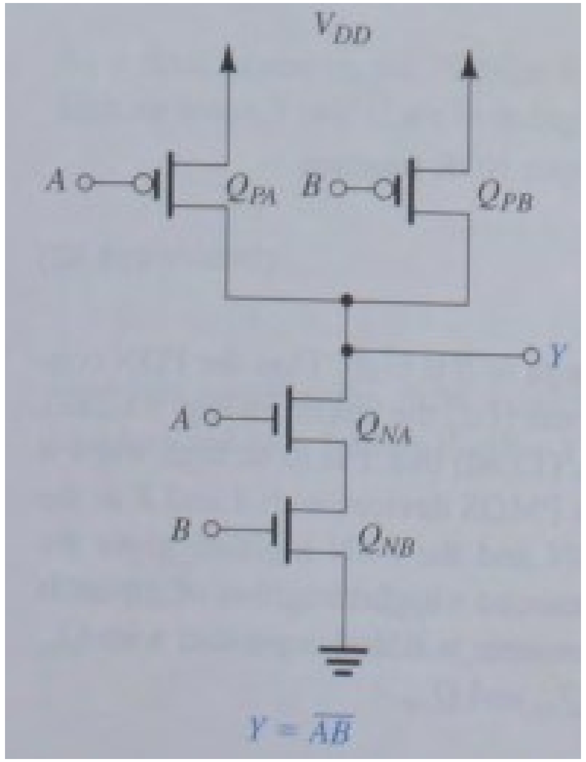
Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

Applications of CMOS: Two-Input NOR Gate



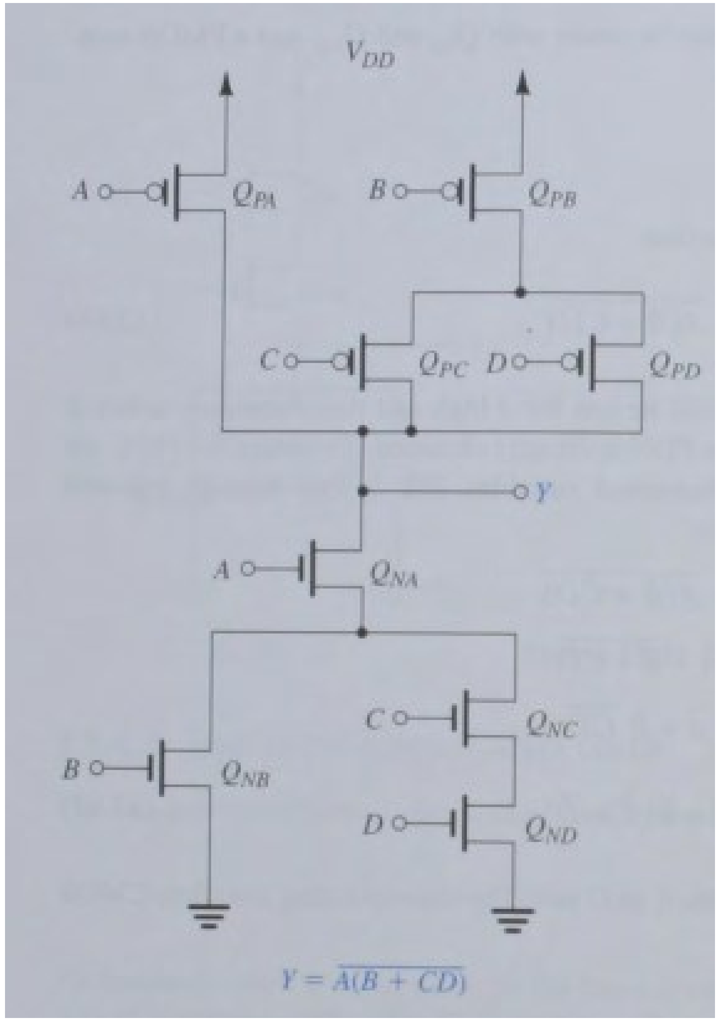
Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

Applications of CMOS: Two-Input NAND Gate



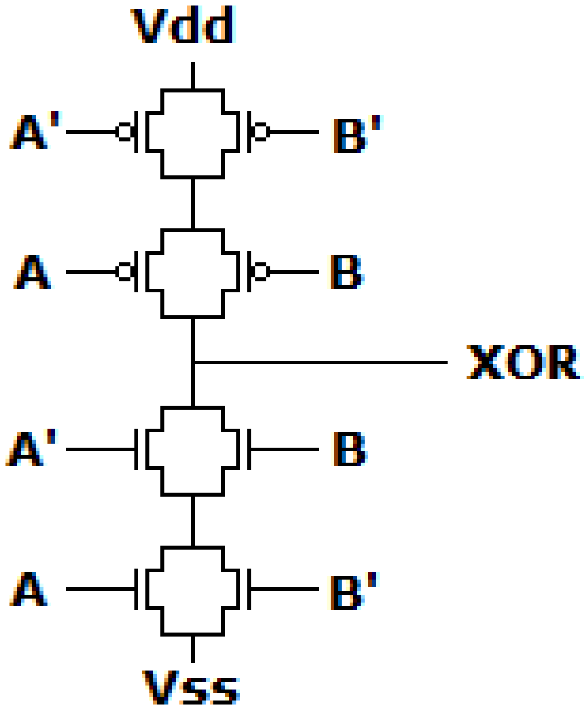
Sedra, Smith, Carusone & Gaudet: “Microelectronic Circuits”, Oxford University Press, 2019.

Applications of CMOS: An Arbitrary Gate



Sedra, Smith, Carusone & Gaudet: "Microelectronic Circuits", Oxford University Press, 2019.

Applications of CMOS: The XOR Gate



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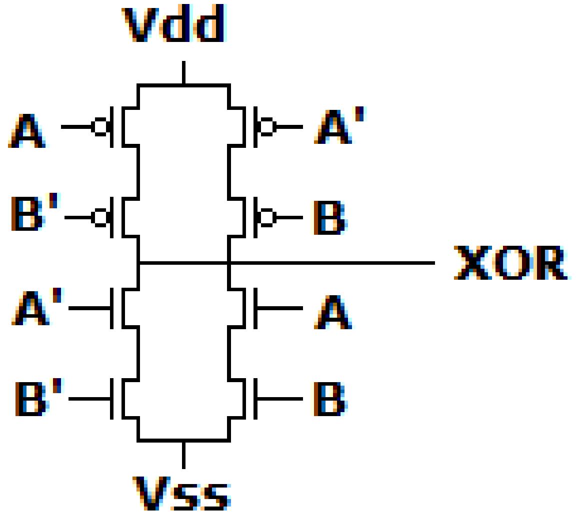

2

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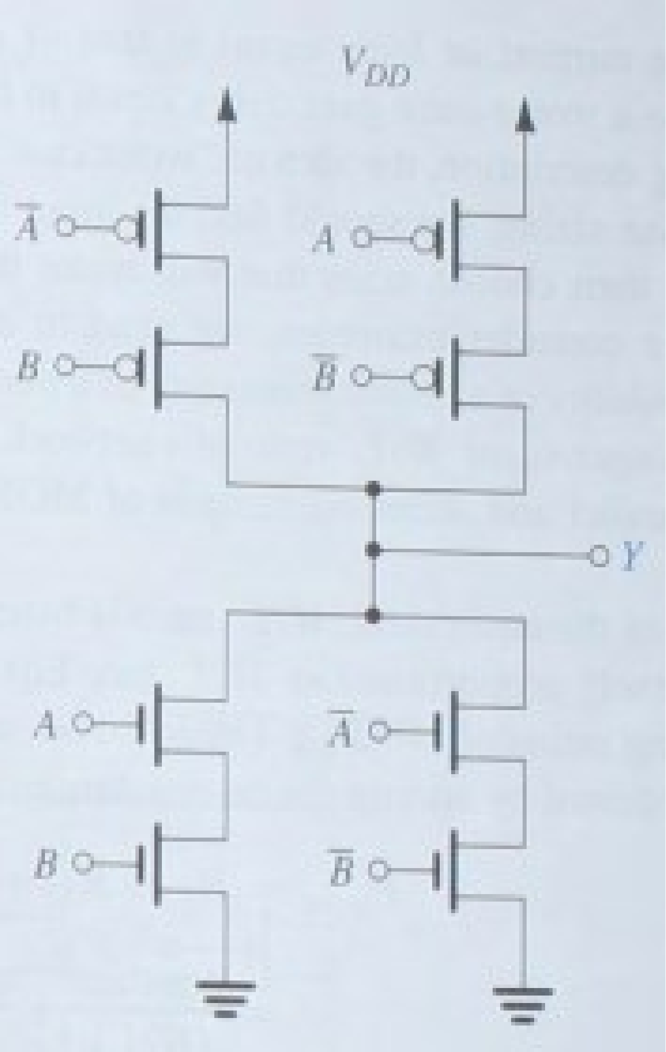
Applications of CMOS: The XOR Gate



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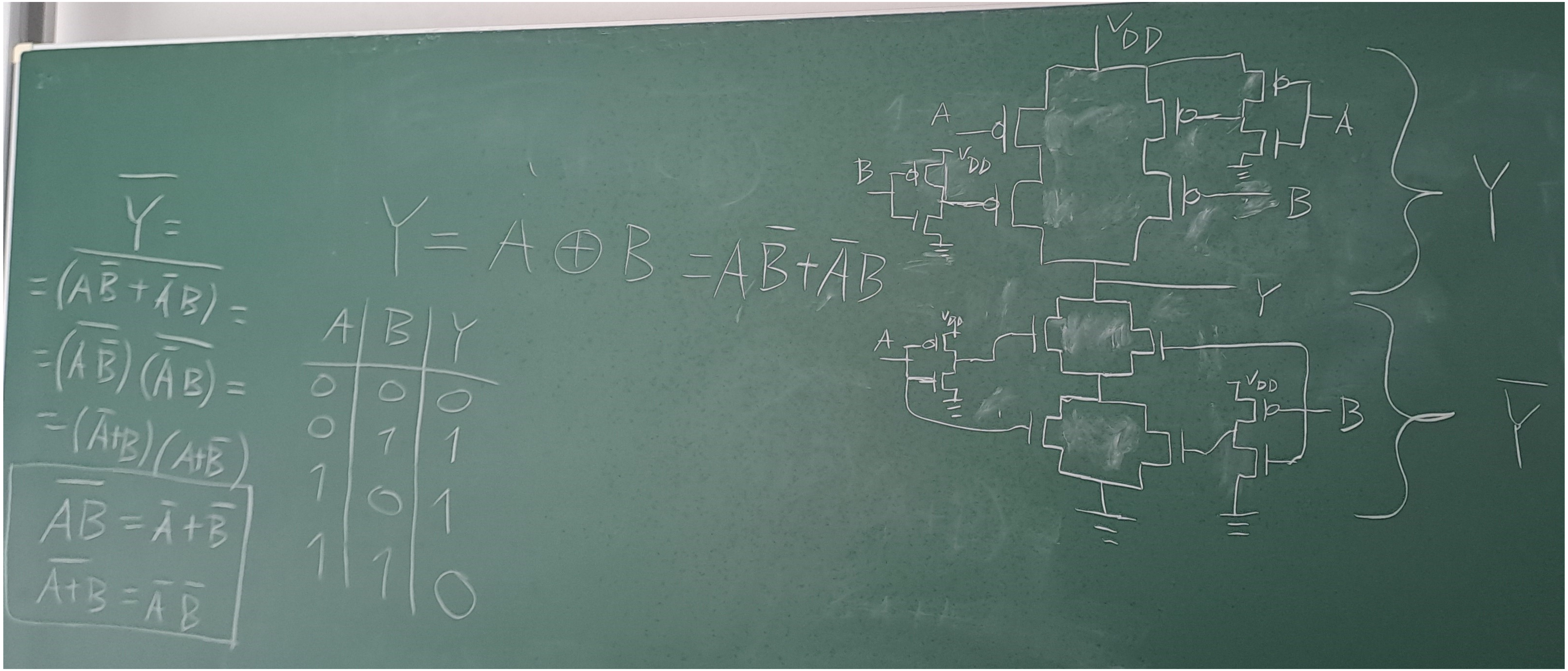
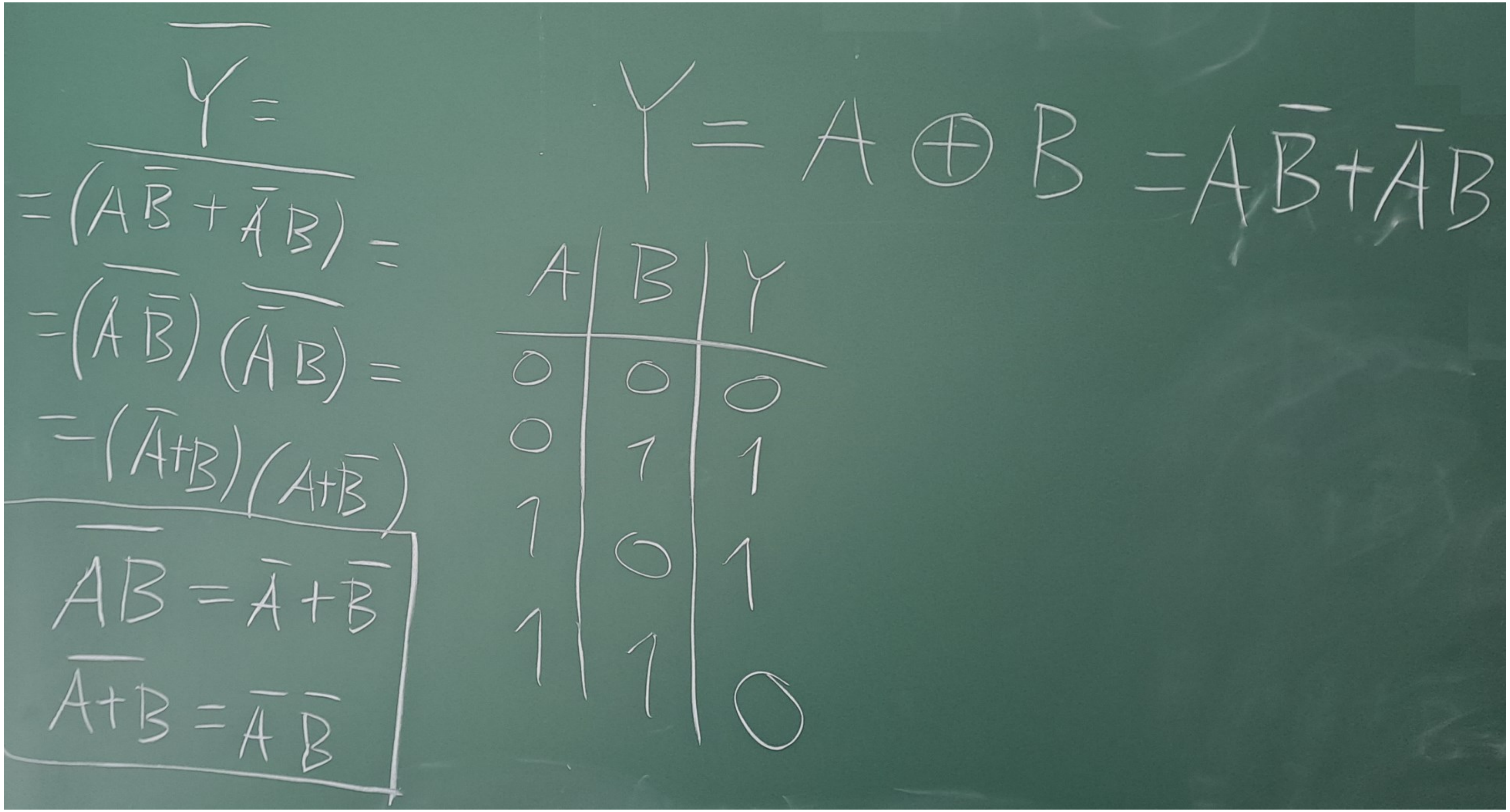
Applications of CMOS: The XOR Gate



Applications of CMOS: The XOR Gate

University of Passau
Faculty of Computer Science and Mathematics



Applications of CMOS: The XOR Gate